Contribution Journal

My main contribution was writing the missing Python code for the construction, training, and visualization of the polynomial regression models and their learning curves. To do this, I first built pipelines using Scikit-Learn to iterate over degrees 1, 4, and 15, and to plot their predictions against generated data. Then, I wrote the plot_learning_curve function for both the degree 1 (underfit) and degree 15 (overfit) cases, with cross-validation set to 5. I then supplemented this code by writing analytical answers for the questions, explaining how complexity influences fit, the interpretation of the trends in the learning curves (i.e., bias/variance trade-off), and how this influences the choice of model.